# Stepwise notes for using PSG

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## Design of Gears

| | SPUR | HELICAL | BEVEL |
|---|---|---|---|
| Finding min of teeth to avoid interference formula | 8.1 | 8.1 | 8.1 |
| Deciding No of Teeth | - | 8.51 | 8.52 |
| Calculating Lewis form factor | 8.50 | 8.50 | 8.50 |
| Material selection | 1.9 or use 8.5 (same as helical) | 8.5 | 8.5 |
| If 1.9 material selected –<br>To calculate design bending stress<br>Design contact compressive stress | <br>8.18 , 8.19<br>8.16 | - | - |
| To find out weaker element | 8.50 | 8.51 | 8.52 |
| To find module based on bending criteria | 8.13A | 8.13A | 8.13A |
| Checking for surface contact compressive stress | 8.13 | 8.13 | 8.13<br>(For R 8.38) |
| For dynamic load, To calculate gear tooth strength Fs | 8.50 | 8.51 | 8.52 |
| To calculate dynamic load ( by any method) Fd | 8.50,8.51 | 8.50,8.51 | 8.52 |
| To calculate tangential force Ft | 8.57 | 8.57 | 8.57 |
| To check for Wear Fw | 8.51 | 8.51 | - |
| Condition for checking | Fs>Fd<br>Fw>Fd | Fs>Fd<br>Fw>Fd | Fs>Fd |

| | Worm and WW |
|---|---|
| Finding min of teeth to avoid interference formula | 8.1 |
| Deciding No of starts and no of Teeth | 8.52 |
| Lead angle | 8.44 |
| Helix angles | 8.52 |
| Calculating Lewis form factor | 8.50 |
| Material selection | 8.45 |
| To find out weaker element | 8.52 |
| To find module based on surface contact comp stress | 8.44,8.43 |
| Checking for bending stress | 8.44 |
| For dynamic load, To calculate gear tooth strength Fs | 8.52<br>For b 8.48<br>For d1 8.43 with x=0 |
| To calculate dynamic load ( by any method) Fd | 8.52<br>For Velocity d2 8.43 |
| To calculate tangential force Ft | 8.57 |
| To check for Wear Fw | 8.52 |
| Condition for checking | Fs>Fd<br>Fw>Fd |

## Rolling Contact Bearings

| | DGBB | Roller |
|---|---|---|
| Equivalent Load | 4.2 | 4.2 |
| Dynamic capacity C & L10 | 4.2 | 4.2 |
| Probability of survival L | 4.2 | 4.2 |
| Cyclic load | 4.2 | - |
| X & Y factor for Equivalent load | 4.4 | - |
| Bearing available series | 4.12-4.15 | 4.22-4.36 |

## Sliding Contact Bearings

| | $360^o$ / full Journal | $180^o$ | $120^o$ |
|---|---|---|---|
| Eccentricity e | Given in question or use formula 7.33 | Same as $360^o$ | Same as $360^o$ |
| Eccentricity Factor/ Attitude Є | Given in question or use formula 7.33 | Same as $360^o$ | Same as $360^o$ |
| Min oil film thickness h0 | Given in question or use formula | Same as $360^o$ | Same as $360^o$ |
| Diametrical clearance C | Given in question or use fit 3.4 H8e8 | Same as $360^o$ | Same as $360^o$ |
| Bearing Pressure P | P=W/LxD | Same as $360^o$ | Same as $360^o$ |
| Assume 2 h0 /c | 7.36 When e or Є is not given | 7.37 When e or Є is not given | 7.38 When e or Є is not given |
| To find viscosity Z, Sommerfeld no S | 7.36 | 7.37 | 7.38 |
| Sommerfeld no formula | 7.34 | 7.34 | 7.34 |
| To find µ coefficient of friction, µD/C | 7.36 | 7.37 | 7.38 |
| Frictional power loss | Hg from 7.34 | Hg from 7.34 | Hg from 7.34 |
| Oil flow rate q, 4q/DCn’L where (n’=n/60) | 7.36 | 7.37 | 7.38 |
| Side leakage qs | 7.36 | 7.37 | 7.38 |
| Operating temp of oil to ρC’delta to/P | 7.36 | 7.37 | 7.38 |
| Selection of oil | 7.41 | 7.41 | 7.41 |

## Design of belts

| | Flat Belt | V Belt |
|---|---|---|
| Pulley Diameters Driver d | Assume v=17.5 to 22.5 m/s (7.53)<br>Select standard value of d from 7.54 | Assume v=17.5 to 22.5 m/s (7.53)<br>Select standard value of d from 7.54 |
| Driven pulley dia D | 7.61<br>Select standard value of D from 7.54 | 7.61<br>Select standard value of D from 7.54 |
| To find lap angle | $\alpha = \sin^{-1}$(D-d/2C) | $\alpha = \sin^{-1}$(D-d/2C) |
| Length of Belt | 7.53 | Approx Centre Dist = C/D for D/d |

| | | |
|---|---|---|
| | | from 7.61<br>Length 7.53 |
| Finding $T_1$ and $T_2$ | $T_1/T_2 = e^{\mu\theta}$<br>$P=(T_1 - T_2)V$ | $T_1/T_2 = e^{\mu\theta/\sin\beta}$ $\beta = 20^o$<br>$P=(T_1 - T_2)V$ |
| Life of Belt | $n = V/L \times L_h \times n_p \times 3600$<br><br>$n/10^7 = (\sigma_{-1}/\sigma_{max})^m$ m=6 | $n = V/L \times L_h \times n_p \times 3600$<br><br>$n/10^7 = (\sigma_{-1}/\sigma_{max})^m$ m=8 |
| $\sigma_{max}$ | $(T_1/bt) + (Et/d) + (\rho v^2 \times 10^{-6})$<br><br>Assume<br>P = 1200 kg/m3<br>E = 100 N / mm2 | $(T_1/n_b \times a) + (Et/d) + (\rho v^2 \times 10^{-6})$<br><br>E = 100 N / mm2<br>P = mass / volm = mass per length / area<br><br>Width of Pulley<br>$L = (n_b - 1) e + 2f$ |
| Design of driver shaft | $T = (T_1 - T_2) d/2$<br>$M = (T_1 + T_2) xW$<br>W = b+13 (7.54)<br>Find Teq and using τ equation find ds1 | $T = (T_1 - T_2) d/2$<br>$M = (T_1 + T_2) xW$<br>W = b+13 (7.54)<br>Find Teq and using τ equation find ds1 |
| Design of driven shaft | $T = (T_1 - T_2) D/2$<br>$M = (T_1 + T_2) xW$<br>W = b+13 (7.54)<br>Find Teq and using τ equation find ds2 | $T = (T_1 - T_2) D/2$<br>$M = (T_1 + T_2) xW$<br>W = b+13 (7.54)<br>Find Teq and using τ equation find ds2 |
| When **life of belt not given**<br><br>To find width and thickness of belt | Selecting 'Dunlop' 878 g (7.52)<br>Load rating (7.54)<br>mm plies of belt 7.54<br>from 7.52 Assume 3 ply find b and check whether it is available or else assume 4 ply, 5 ply and so on<br>thickness = t = no of ply x 1 mm + 2mm | Calculating $n_b$<br>PSG 7.70<br>For $F_a$ 7.69<br>$F_c$ 7.60<br>$F_d$ 7.68<br><br>To find out KW rating from 7.65<br>For V and de = d X Fb<br>$F_b$ 7.62 |

## Design of Clutches

| | Multi plate Clutch | Cone Clutch |
|---|---|---|
| Service Factor | 7.89 , 7.90 , 7.91 | 7.89 , 7.90 , 7.91 |
| Design Torque | 7.89 | 7.89 |
| Input Shaft Design | Using τ equation | Using τ equation |
| Output splined shaft design | Selecting from 5.30 | Selecting from 5.30 |
| Friction Plate Design | 7.89,7.90 μ and Temp 7.97 $i_{min}$ 7.89<br>No of driving and driven plates 7.90 | Cup and cone design :<br>7.89,7.90 μ and Temp 7.97<br>Assume i = 1 and find b using I min formula<br>R min = Rm- b/2 sinα |

| | | |
|---|---|---|
| | | R max = Rm + b/2 sinα |
| Pressure plate design | Same dimensions as friction plate | - |
| Heat dissipation | Power loss = [Mt] x Θs x n/3600<br>$hA(T_h-T_a) + \sigma_s A (Th^4- Ta^4)$<br>Ta = 30 $^oC$<br>h = 15 w/m2k<br>$\sigma_s = 5.6 \times 10^{-8}$ | Power loss = [Mt] x Θs x n/3600<br>$hA(T_h-T_a) + \sigma_s A (Th^4- Ta^4)$<br>Ta = 30 $^oC$<br>h = 15 w/m2k<br>$\sigma_s = 5.6 \times 10^{-8}$ |
| Lever Design | Pact x i (whole no) = P allowable x I (Value in points) 7.90<br>Axial Force 7.90<br>Force on each lever Q' 7.90<br>Q" 7.90<br>Lever pin dia d1 7.90<br>Smaller arm length a= ($r_{max}$ –d) + d/2 + 5mm<br>h2 7.90<br>to check lever pin for bearing failure 7.91 | Not Applicable |
| Spring Design | Assuming No of Springs as no of levers<br>Pmin = Q'<br>Pmax = 1.25 P min<br>Pmax = P min + qy 7.102<br>Assume y = 5 mm<br>Find q<br>Assume C = 6 ,<br>Find Ks and find d using τ equation from 7.100<br>Find no of turns n from q equation 7.100 | Normal force acting on friction surfaces<br>$F_n = P_m \times 2\pi r_m \times b$<br>Axial force on friction liner<br>$F_a = F_n \times \sin\alpha$<br>**Min force to engage the clutch**<br>**Pmin = Fa**<br>**Max. force to disengage the clutch**<br>**Pmax = 1.2 Pmin**<br>Pmax = P min + qy 7.102<br>Assume y, Find q<br>Assume C = 6 ,<br>Find Ks and find d using τ equation from 7.100<br>Find no of turns n from q equation 7.100 |

# Design of Roller Chains

| | |
|---|---|
| Service Factor | 7.76,7.77 |
| Selection of pitch | 7.74<br>> 500 rpm 9.525mm<br>200-500rpm 12.7mm<br><200rpm 15.875mm |
| No of teeth<br>Driver sprocket $Z_1$<br>Driven sprocket $Z_2$ | <br>7.74<br>7.74 |
| Pitch circle dia of<br>Driver sprocket $d_1$<br>Driven sprocket $d_2$ | <br>7.78<br>7.78 |

| | |
|---|---|
| Velocity of chain | Лd$_1$N$_1$/60 |
| Breaking load calc | 7.77 here N is power |
| Selection of chain based on breaking load and pitch | 7.71-7.72 |
| Checking for bearing failure | 7.77 here N is power |
| Length of chain | 7.75 $a_p$ = 40 |
| Final centre distance | 7.75 |
| Actual factor of safety | 7.78 put power in KW and Q should be of selected chain not calculated one |

## Design of Cam and Follower

| | |
|---|---|
| Max Pressure angle (α ) | Either given or Assume α1 = $25^o$ and α2 = $30^o$ |
| Displacement dia , velocity dia and acceleration dia | If cycloidal motion is given check disp., velocity and acceleration at 5 points (0 , $\beta_1/4$, $\beta_1/2$, $3\beta_1/4$ and $\beta_1$) and for SHM check at three points ($\beta_2$, $\beta_2/2$ and 0) PSG 7.110 put β in radians |
| Calculation for prime circle radius Rp | Radius of roller Rr either mentioned in Q or assume 10 mm<br>Use formula of tan α from 7.113<br>$Y_\Theta$ is that displacement at which velocity is max<br>dY/dΘ = (Vmax/ω)<br>find Rp for rise/ forward and Rp for fall/return<br>whichever is higher select that |
| Base circle radius Rb | Rp-Rr |
| Calculation of minimum radius of curvature for pitch curve $\rho_{kmin}$ | 7.114<br>Take Y = Y max and $d^2y/dt^2$ = max negative acceleration<br>dy/dt = velocity at point where max negative acc occur |
| Radius od curvature of cam profile $\rho_{cmin}$ | $\rho_{kmin}$ - Rr |
| Force analysis | Make table for all the points<br>F = Fe+ Wt of Follower + Finertia<br>=Fe+($m_f$g) +($m_f$a)<br>Here Fe will vary for rise and Fall and acceleration a will vary at each point |
| Spring force | If at any point total force comes out to be negative, cam will loose contact with follower<br>Hence add Spring with Pmax = force value + 30N<br>Assume initial deflection in spring 2 mm<br>Total deflection = 2mm + Ymax<br>Spring Stiffness (K) = Pmax / total deflection |
| Recalculation force analysis | F + Kδ<br>δ = 2mm + disp at that point<br>find out point at which max force occur |
| Calculation of cam width b | 7.115 $\sigma_c$<br>Calculate $P_n$ using $P_n$cosα = Fmax (use that α where point of Fmax occur) |

| | |
|---|---|
| Calculation of max cam shaft torque Mt | 7.116 Fmax.Vmax/ω |
| Design of roller pin | Length of Pin = b + 10 mm<br>Max bending moment = M = WL/4 = Fmax.L/4<br>Find d using τ equation<br>Check for double shear |
| Follower Stem | Length of Stem = 2.5 x max lift<br>Slenderness Ratio = Ls/radius of pin<br>Checking for buckling using Johnsons formula 6.8 Pc should be > Fmax to avoid buckling |
| Design of Spring | Assume y = 5 mm<br>Find q<br>Assume C = 6 ,<br>Find Ks and find d using τ equation from 7.100<br>Find no of turns n from q equation 7.100 |